ANSWER SHEET DAC 2026

PRELIMINARY ROUND

Data Analysis Competition 2026

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CHAPTER I: Introduction

A. Background

Healthcare insurance administrators must process legitimate claims promptly while protecting limited public resources from fraud, abuse, and other irregular claim patterns. In practice, however, manual investigation cannot be applied to every submitted claim. Fraud-detection systems are therefore most useful when they help investigators concentrate limited review capacity on claims with the highest expected risk rather than attempting to replace human judgment altogether 1[1], [2].

This operational constraint is central to the DAC 2026 case. The Nusantara Health Protection Agency (NHPA) can manually review at most 5% of incoming claims. Consequently, the problem is not simply to classify each claim as fraudulent or legitimate. The system must produce a reliable ranking in which fraudulent claims are concentrated near the top of the list, because only the highest-ranked 5% will enter the audit portfolio. A missed high-risk claim may expose the insurance fund to avoidable loss, whereas an unnecessary audit consumes investigator capacity and may delay or burden a legitimate policyholder. The resulting decision therefore involves a direct trade-off between fraud capture and unnecessary review.

Developing such a system also requires careful treatment of the competition data. Although the supplied dataset contains no missing values, several characteristics can affect the validity of the analysis. Administrative variables such as kdkc and dati2 are codes rather than continuous quantities and must preserve their categorical meaning during preprocessing. The development data also contain repeated claims with identical observed feature signatures, including 1,737 signature groups in which otherwise identical records have conflicting labels. Such cases indicate that the observed variables do not perfectly determine the audit outcome and make ordinary row-wise validation potentially optimistic if identical claim profiles appear in both training and validation partitions.

A separate forensic inspection identified a storage-order artifact in the released training file, where labels occur in contiguous blocks. Because row position and claim_id are not information available for legitimate fraud inference, neither is used as a predictive feature or tie-breaking signal. This distinction between genuine claim information and accidental dataset structure is important for preventing information leakage and ensuring that validation measures the model rather than the organization of the source file. In addition, the approximately balanced fraud proportion in the development dataset should be regarded as a property of the competition sample rather than an estimate of real-world fraud prevalence. Probability estimates may therefore require prospective recalibration before operational deployment.

To address these challenges, this study develops an Integrated Claim Risk and Audit Allocation System centered on a homogeneous CatBoost modeling pipeline.

CatBoost is suitable for this setting because it can directly represent heterogeneous categorical variables while capturing nonlinear interactions among demographic, geographic, facility, diagnosis, procedure, severity, and claim-burden characteristics [3]. The model is complemented by semantically informed feature construction, while the validation procedure uses five-fold Stratified Group Cross-Validation based on exact cleaned-feature signatures so that identical observed claim profiles cannot be shared between the training and validation sides of a fold.

Model development is evaluated from more than one perspective. The official NormalizedRecall@5% metric measures whether fraud is successfully concentrated within the available audit capacity, while analyses at 3%, 5%, and 7% translate model ranking into operational workload. Average Precision and probability calibration are examined as supporting evidence of ranking quality and risk-score reliability. Model drivers and individual high-risk predictions are analyzed to support investigator interpretation, while legitimate-claim audit rates are compared across gender and the official age groups to identify possible differences in audit exposure. These disparities are treated as monitoring signals rather than evidence of discrimination by themselves [4].

Formally, let $D = \{(x_i, y_i)\}_{i=1}^{n}$ denote the available claims, where x_i contains the permitted claim information and $y_i \in \{0, 1\}$ indicates the observed legitimate or fraud label. The model learns a scoring function $f_\theta(x_i) = \hat{p}_i$, $0 \leq \hat{p}_i \leq 1$, where a larger $\hat{p}_i$ represents higher estimated fraud risk. For N incoming claims, only the $K = \lfloor 0.05N \rfloor$ highest-ranked claims are allocated to the primary audit portfolio. The analytical objective is therefore to concentrate as much fraud as possible within this limited portfolio while maintaining reliable risk estimates, minimizing unnecessary audits, protecting policyholders, and preserving meaningful human oversight.

B. Research Questions

Based on the problem and limitations described above, this study is guided by the following research questions:

- **RQ1:** What data-quality characteristics, repeated claim patterns, unusual observations, proxy variables, and potential leakage risks must be addressed to obtain a valid fraud-risk analysis?
- **RQ2:** How effectively can a leakage-aware and interpretable CatBoost model rank fraud risk while maintaining reliable validation and probability estimates?
- **RQ3:** How much fraud can be captured under 3%, 5%, and 7% audit capacities, and how much legitimate-claim review can be avoided relative to simple and random audit strategies?

- **RQ4:** Do legitimate policyholders from different gender and age groups experience materially different audit-selection rates, and how should any observed disparities be interpreted?
- **RQ5:** How should the resulting fraud-risk scores be incorporated into a human-centered operational process that supports explanation, override, escalation, appeal, monitoring, recalibration, and retraining?

Together, these questions frame fraud detection not as an isolated prediction problem, but as a constrained decision-support problem involving data reliability, risk ranking, audit allocation, policyholder protection, and operational governance.

CHAPTER II: Theoretical Framework

A. Fixed-Budget Ranking Metrics

Fraud detection under limited investigation capacity differs from conventional binary classification. In ordinary classification, a threshold is selected to assign each observation to a positive or negative class. In the NHPA setting, however, the number of investigations is predetermined: only the highest-risk $K = \lfloor qN \rfloor$ claims can be audited, where q denotes the available audit capacity. The principal objective is therefore to produce a risk ranking that concentrates fraudulent claims within this limited portfolio rather than to maximize classification accuracy at an arbitrary threshold [6], [7].

Let P denote the total number of fraudulent claims, K the available audit slots, and, and TP the fraudulent claims contained in the top-K portfolio. Recall at audit capacity q is $Recall@q = \frac{TP}{P}$. For the official 5% budget, DAC 2026 normalizes this quantity by the maximum achievable recall: $NormalizedRecall@5\% = \frac{TP}{P}$. An important consequence occurs when $P > K$, as in the development dataset. In this case, $NormalizedRecall@5\% = \frac{TP/P}{K/P} = \frac{TP}{K}$, so the metric is equivalent to the fraud precision of the fixed audit portfolio. This gives the competition metric a direct operational interpretation: increasing the score means increasing the proportion of the limited audit slots that contain fraudulent claims.

Because operational capacity may change, a useful system should not be evaluated at a single cutoff alone. Accordingly, performance is also examined at 3%, 5%, and 7% audit capacities. These views quantify the trade-off between additional fraud captured and additional legitimate claims subjected to investigation.

B. Grouped Validation

Reliable fraud modeling depends not only on model choice but also on whether validation represents genuinely unseen observations. Data leakage occurs when information that would not legitimately be available at prediction time, or

information derived from the validation outcome itself, enters model development and produces overly optimistic performance estimates [8].

This issue becomes particularly important when observations are statistically dependent. Standard random cross-validation assumes observations are sufficiently independent, whereas repeated, spatial, temporal, or hierarchical structures can cause related records to appear on both sides of a split [9]. Model selection performed repeatedly against the same validation information may further introduce selection bias even when no explicit target leakage is present [10].

The dataset used in this study contains repeated exact feature signatures. Therefore, the appropriate validation unit is not an individual row but an exact observed claim profile. A grouped cross-validation design assigns all records with the same cleaned-feature signature to the same fold while maintaining approximately balanced class distributions. This prevents an identical representation from being learned in the training partition and evaluated again in its validation partition.

The same principle applies to engineered representations. Any data-dependent statistic, such as category frequencies, must be estimated using the training portion of a fold and subsequently applied to the held-out portion. Furthermore, identifiers and source-row positions should never act as predictive variables. These safeguards allow observed validation performance to reflect relationships among legitimate claim characteristics rather than accidental properties of dataset construction.

C. Gradient Boosting and CatBoost

Gradient Boosting builds a predictive model sequentially by combining weak learners, typically decision trees, where each new learner is trained to reduce the errors of the current ensemble. The prediction after iteration m can be represented as $Fm(x) = Fm - 1(x) + \eta hm(x)$, where $hm(x)$ denotes the new tree and η controls its contribution. This additive structure enables boosted trees to capture nonlinear effects and interactions without requiring explicit specification of every possible relationship between variables.

Healthcare claims are particularly suited to tabular models because they contain mixed numerical and categorical information, including administrative locations, facility types, service categories, diagnoses, procedures, and severity indicators. Several of these variables are identifiers or semantic codes rather than meaningful numerical magnitudes.

CatBoost extends gradient boosting by handling categorical variables directly and constructing categorical statistics through an ordered procedure designed to reduce target leakage and prediction shift [3]. This is relevant to variables such as kdkc, dati2, typeppk, cmg, and diagprimer, whose relationships with fraud may be nonlinear and dependent on interactions with other claim characteristics.

The selected system therefore uses CatBoost as a single model family, complemented by claim-level burden, concentration, density, and limited interaction features. The same model specification is trained across validation folds and its test predictions are averaged. This retains the variance-reduction benefit of cross-validation inference while avoiding a heterogeneous stacking architecture that would add complexity to explanation, maintenance, and governance.

D. Probability Reliability and Audit Allocation

Ranking quality and probability quality are related but distinct objectives. A model can correctly order high-risk claims while producing poorly calibrated probabilities, and conversely a well-calibrated model may still fail to rank the most consequential claims near the audit boundary. Because the submitted output is called fraud_probability, both aspects should be assessed.

Probability calibration describes the agreement between predicted risks and observed event frequencies. One common measure is the Brier Score [11]: $Brier = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2$, where lower values indicate more accurate probabilistic forecasts. Calibration is particularly relevant when scores are shown to human reviewers or used to compare risk across portfolios [12].

Post-hoc calibration methods such as logistic or beta calibration can adjust probability scale while preserving much of the underlying ranking [12], [13]. However, calibration must be handled carefully in a fixed-budget application. A strictly monotonic transformation applied consistently to all observations preserves their order, whereas fold-specific transformations or calibrators that create probability plateaus can alter the global top-K portfolio.

For this reason, this study separates ranking evaluation from calibration evaluation. Calibration is considered beneficial only when it improves probability reliability without materially degrading audit ordering or creating excessive ties around the selection threshold. When an exact tie occurs in internal evaluation, a deterministic label-neutral rule is used solely to ensure reproducibility; claim_id remains excluded from the predictive feature set.

A further limitation is that calibration estimated on the competition sample reflects its approximately balanced development distribution. It should not automatically be interpreted as real-world NHPA fraud incidence. If deployment prevalence differs substantially, prospective recalibration using operational outcomes would be required.

E. Interpretability and Policyholder Protection

A high fraud-risk score should support investigation rather than function as evidence of wrongdoing. Consequently, model interpretation must explain which observed characteristics contributed to a prediction without treating statistical association as proof of fraud causality.

Global interpretation can be obtained through CatBoost feature importance and held-out permutation importance, while local explanations can be generated using SHAP values [14]. SHAP represents a prediction as contributions from individual features relative to a reference prediction, allowing investigators to understand why a particular claim received a comparatively high or low score. Because correlated variables can redistribute importance among one another, attribution results should be interpreted as model-behavior explanations rather than causal effects.

Policyholder protection requires a separate analysis because strong average ranking performance does not guarantee equal audit exposure. Following the competition requirement, legitimate claims are grouped by gender and by age categories $0-17$, $18-39$, $40-59$, and $60+$. For a protected group g, the legitimate-claim audit rate is $FPRg = \frac{legitimate\ claims\ in\ group\ g\ selected\ for\ audit}{all\ legitimate\ claims\ in\ group\ g}$. Differences between group-specific audit rates indicate differential exposure to unnecessary investigation. Such differences should be reported with appropriate uncertainty and context, but they do not by themselves demonstrate discrimination [4]. They may reflect clinical composition, geography, service patterns, data quality, model behavior, or combinations of these factors.

Fairness evaluation should therefore complement, rather than replace, model interpretation. Geographic and facility variables can carry legitimate structural information while simultaneously acting as proxies for population characteristics, making continued subgroup monitoring important even when initial demographic gaps appear small.

F. Robustness, Drift, and Human Oversight

Cross-validation estimates performance under assumptions about how future claims resemble the development data. Strong in-distribution performance does not guarantee transportability to new regions, providers, or fraud patterns. Structured stress testing is therefore useful for identifying contexts in which the model may become less reliable.

In this study, geographic transportability is examined separately by evaluating claims from administrative codes excluded from model training. This stress test is not used to select the final model; instead, it measures sensitivity to distributional changes. More generally, deployment systems should monitor changes in feature distributions, score distributions, audit yield, calibration, and subgroup audit rates because fraud behavior and healthcare utilization patterns may evolve over time [15].

Drift detection alone is insufficient for safe operational use. A high-risk alert should be accompanied by relevant claim information and model explanations, after which trained investigators determine whether additional evidence is required. Human reviewers should be able to override model prioritization when clinically or administratively justified, while disputed decisions require escalation

and appeal mechanisms. Model outputs should never autonomously reject claims [5].

The same feedback process supports model maintenance. Confirmed investigation outcomes, override patterns, appeals, calibration changes, and emerging subgroup disparities can indicate when recalibration or retraining is warranted. This creates a feedback loop in which the model assists human decision-makers while operational evidence is subsequently used to monitor the model itself.

G. Related Work

Machine learning has been widely explored for detecting fraud and abuse in healthcare claims, including supervised classification, anomaly detection, and hybrid analytical approaches. Reviews of the field emphasize that healthcare fraud data commonly contain heterogeneous administrative and clinical variables, uncertain labels, evolving fraud mechanisms, and substantial differences between datasets, making validation design as important as algorithm selection [1].

A particularly relevant example is the World Health Organization's study of machine learning within the Philippine national health-insurance claims process, which discusses the potential of predictive analytics to support fraud investigation while emphasizing operational integration and limitations of available claims data [2]. More broadly, healthcare AI governance literature stresses that predictive performance alone is insufficient when systems affect access to healthcare or administrative review; interpretability, human oversight, monitoring, and attention to group-level effects remain necessary [4], [5].

The present study extends these principles to a fixed-capacity audit allocation problem. Rather than claiming a universally optimal fraud classifier, it combines leakage-aware validation, categorical gradient boosting, probability assessment, audit-capacity analysis, policyholder protection, and human-centered governance within the specific data and operational constraints defined by DAC 2026.

CHAPTER III: Analytical Steps

The analytical workflow transforms the provided claim data into a validated fraud-risk ranking and operational audit portfolio. It covers data integrity assessment, semantic-preserving preprocessing, feature construction, leakage-aware validation, model comparison, probability assessment, audit allocation, policyholder-protection analysis, interpretability, and robustness evaluation, as summarized in Figure 3.1.

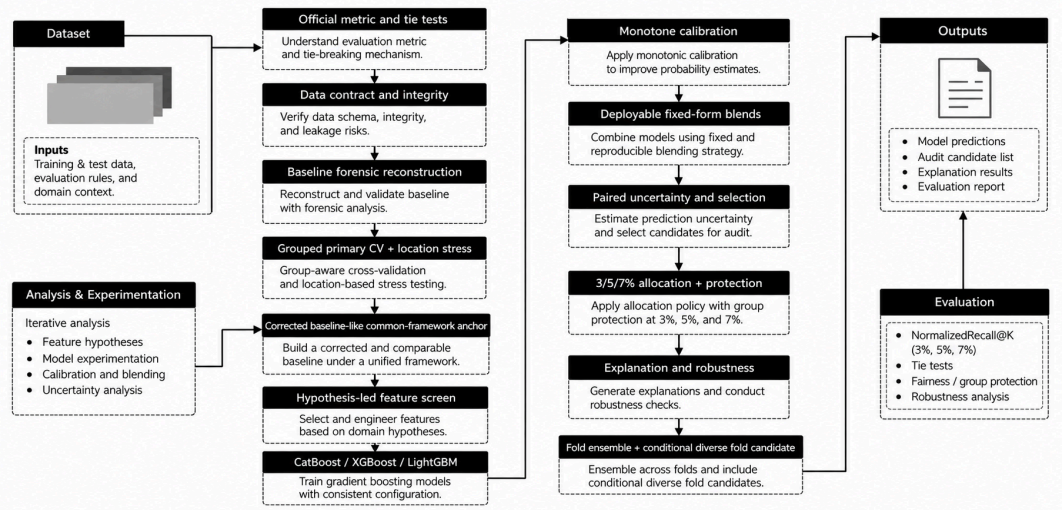


Figure 3.1. Analytical workflow of the claim-risk and audit-allocation system

A. Data Preparation and Integrity Assessment

Data types are defined according to semantic meaning rather than stored representation. Administrative codes such as kdkc and dati2 are retained as strings to preserve leading zeros, while typeppk, jkpst, jnspelsep, cmg, severitylevel, and diagprimer are treated as categorical variables. Whitespace is normalized, numerical count variables remain numerical, and zeros in diagnosis and procedure counts are retained as observed values rather than missing data.

The integrity audit examines missingness, unusual numerical values, categorical cardinality, duplicate and repeated feature signatures, conflicting labels, train-test overlap and shift, and source-order artifacts. claim_id and source-row position are excluded from all predictive features. Associations between variables and fraud are used descriptively to identify patterns and possible proxies, but are not interpreted as evidence of fraud causality.

B. Feature Construction

The original claim variables are retained and supplemented with compact, hypothesis-driven feature families representing utilization intensity, clinical-code structure, and administrative context.

Table 3.1. Feature families and analytical purpose

Feature Family	Examples	Analytical Purpose
Demographic and stay structure	age_band, is_pediatric, is_older_adult, log1p_los, zero_los	Represent age segments and nonlinear length-of-stay behavior
Claim burden	diagnosis/procedure totals, nonzero group counts, total_coded_burden	Measure overall diagnosis and procedure intensity

Burden density	burden_per_day, procedure_per_day, diagnosis_per_day, procedure_to_diagnosis	Distinguish similar claim volumes occurring over different lengths of stay
Code concentration	diagnosis/procedure entropy and maximum share	Distinguish concentrated coding patterns from diffuse ones
Administrative hierarchy	kdkc_prefix2, dati2_prefix2	Preserve coarse structure within anonymized administrative codes
Structured interactions	diagnosis × CMG, diagnosis × service, diagnosis × facility, CMG × facility, KDKC × DATI2	Capture selected relationships among clinical and administrative characteristics
Fold-safe rarity	frequencies of selected geographic and clinical combinations	Measure how common or unusual a pattern is without using validation information

For the dx2_* and proc* blocks, counts are summarized through totals, active-group counts, maxima, entropy, and concentration measures. Density features additionally normalize selected quantities by length of stay.

Frequency-based rarity variables are generated fold-safely: frequencies are estimated only from the training portion of each fold and then mapped to validation and test observations. Feature-family ablation is used to verify that engineered representations provide useful evidence without unnecessary expansion.

C. Validation Strategy

Because multiple rows can share an identical observed claim profile, ordinary random splitting can place equivalent signatures on both sides of the validation boundary. The primary evaluation therefore uses five-fold StratifiedGroupKFold, with the exact cleaned feature signature as the grouping key and class balance preserved across folds.

Each claim receives exactly one out-of-fold prediction. The first three folds provide the model-selection perspective, while the remaining two serve as confirmation views that can reject unstable improvements. This structure is used as a robustness safeguard rather than as an independent external test.

A separate unseen-kdkc evaluation is used only as a geographic transportability stress test. It does not determine model selection. All model-development decisions are based on training-data out-of-fold evidence rather than leaderboard-driven claim-level adjustment.

D. Model Development and Selection

Model development follows a controlled comparison rather than unrestricted hyperparameter search. A transparent burden-based model serves as an operational baseline, followed by Logistic Regression, XGBoost, LightGBM, a reference CatBoost model, and the final CatBoost specification. All candidates are evaluated on identical folds and metrics.

CatBoost is selected as the principal architecture because it can directly represent the dataset's high-cardinality categorical variables while learning nonlinear interactions with clinical and utilization features. The selected configuration uses 2,860 boosting iterations, learning rate 0.03, depth 8, and random seed 2127; complete hyperparameters are retained in the reproducible notebook.

No oversampling, undersampling, SMOTE, or class weighting is applied because the development labels are approximately balanced. The principal operational imbalance is instead the 5% audit-capacity constraint and is handled through ranking and top-K evaluation.

Alternative CatBoost seeds and heterogeneous model combinations are screened as stability and complementarity checks. Additional complexity is retained only when improvement is sufficiently large and consistent across folds. When candidates provide practically similar evidence, the simpler model is preferred to preserve interpretability and operational maintainability.

E. Probability, Audit, and Robustness Evaluation

Probability reliability is evaluated separately from ranking performance. Raw CatBoost probabilities are compared with sigmoid, beta, and isotonic calibration using cross-fitted calibration evidence. A calibrator is considered useful only if it improves probability quality without degrading the audit ranking or producing excessive ties near the fixed 5% boundary.

For evaluation-data inference, five fold-specific CatBoost models generate test probabilities, which are averaged to produce the final fraud-risk score. The resulting submission is checked for identifier alignment, unique claims, finite probabilities, required columns, and values bounded within $[0, 1]$.

Policyholder protection is assessed only among legitimate claims across gender and the required age groups: 0–17, 18–39, 40–59, and 60+. Group audit rates are reported with denominators and uncertainty intervals, while disparity gaps summarize differences in unnecessary audit exposure.

Interpretability combines CatBoost feature importance, held-out permutation evidence, and SHAP contributions for representative claims. These methods explain model behavior rather than fraud causality. Robustness is further assessed

through feature-family ablation, seed sensitivity, prediction stability, train-test shift, and the unseen-location stress test.

Together, these analytical steps ensure that the final system is evaluated not only on fraud-ranking performance, but also on leakage control, probability reliability, audit efficiency, policyholder protection, interpretability, and deployment robustness.

These results indicate that observed fraud-risk patterns vary across administrative and facility context, diagnosis-service and diagnosis-CMG combinations, severity structure, and age. These relationships are predictive associations rather than evidence that any characteristic causes fraud.

CHAPTER IV: Analysis of Results

A. Data Characteristics and Limitations

Predictive performance is only meaningful when the validation problem is correctly specified, so the analysis begins with forensics. Training holds 160,174 claims (80,203 fraud, 79,971 legitimate) and evaluation 40,043 claims, with no missing values. The 50.07% training fraud proportion is a competition-sample property, not an operational prevalence estimate.

The data dictionary describes label as a continuous fraud probability, but the training file contains only binary 0/1 values. The target is therefore treated as an observed binary label while the model output remains a continuous risk probability.

Several apparently numerical variables are categorical codes: kdkc has 126 levels (46,210 records with leading zeros), all 160,174 dati2 values preserve leading zeros across 486 levels, and 76,189 raw typeppk values carry trailing whitespace - hence semantic string ingestion and whitespace normalization. Age (max 109) and length of stay (max 592 days) sit within organizer ranges and are retained as unusual observations, not errors.

More consequential are repeated claim structures: after cleaning, 149,015 unique signatures remain, with 6,747 repeated groups (17,906 rows, largest pattern 53x). Among them, 1,737 groups (5,361 rows) carry conflicting labels under identical predictive features, reflecting unobserved information or annotation uncertainty - and justifying one-fold-per-signature validation.

A storage artifact is also visible: fraud labels form one contiguous block followed by legitimate labels, a single transition across the file.

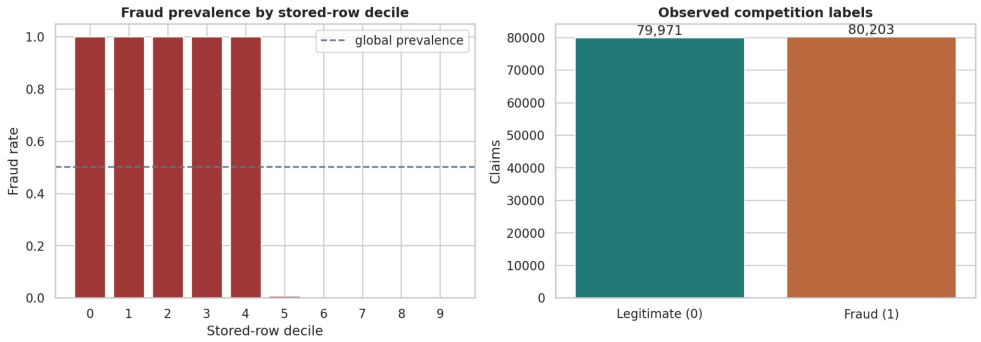


Figure 4.1. Label balance and the storage-order discontinuity identified in the training dataset.

This ordering is unrelated to claim characteristics at audit time, so row position and claim_id are excluded from features. All rarity statistics are computed fold-safely so validation rows never inform their own representation.

Train-test comparison provides a more reassuring picture of the official evaluation cohort. The largest univariate distribution difference is observed for proc39_45, with a shift magnitude of only 0.0133, while cross-validated adversarial validation achieves an AUC of approximately 0.5068. Thus, the supplied train and test cohorts are globally difficult to distinguish from their observed variables. This does not guarantee transportability to genuinely unseen geographic contexts, which is investigated separately in Section F.

B. Fraud-Risk Model Performance and Selection

All candidates share the exact-signature grouped out-of-fold framework (Table 4.1). The burden baseline shows utilization intensity helps but trails boosted models substantially; logistic regression is the interpretable linear reference but misses nonlinear categorical interactions.

Table 4.1. Out-of-fold performance of principal V2 model candidates

Model	NormalizedRecall @5%	Average Precision	Brier Score	Fold SD
Burden baseline	0.604895	0.540779	0.247161	0.007809
Logistic Regression	0.894106	0.746009	0.200328	0.005471
Reference CatBoost	0.951923	0.789910	0.185796	0.002978
XGBoost	0.948427	0.794987	0.183161	0.004206

LightGBM	0.958666	0.803697	0.179844	0.004157
CatBoost, seed 2026	0.968032	0.823088	**0.172217**	**0.001470**
CatBoost, seed 2127	**0.970529**	**0.823183**	0.172383	0.002735

The selected CatBoost V2 seed 2127 achieves a NormalizedRecall@5% of 0.970529, Average Precision of 0.823183, and low cross-fold variability. Relative to the corrected V1-like CatBoost anchor, the selected V2 model increases NormalizedRecall@5% by approximately 0.0186, while paired exact-signature-cluster bootstrap analysis confirms a material improvement in the overall internal evaluation evidence.

More complex combinations were also examined rather than assumed to be superior. Averaging the two CatBoost seeds retained the same 0.970529 NormalizedRecall@5%, while the strongest tested CatBoost-LightGBM logit blend reached 0.971154, an absolute gain of only 0.000625 over the selected model. The additional model did not satisfy the predefined consistency and complementarity safeguards required for promotion. The final system therefore retains the homogeneous CatBoost configuration, prioritizing a simpler model structure when the incremental ranking gain is marginal.

The corresponding selected-model submission obtained a 0.96333 public leaderboard score. This leaderboard result is treated only as external interim feedback and was not used to identify claims, alter scores, or select model components.

Probability Calibration

Probability quality was evaluated separately from ranking. For the selected candidate, raw, sigmoid, and beta calibration produce almost identical results:

Table 4.2. Calibration comparison for the selected CatBoost model

Calibration	NR@5%	Brier Score	Boundary Tie Size
Raw	**0.970529**	0.172383	**1**
Sigmoid	0.970529	0.172381	1
Beta	0.970529	**0.172380**	1
Isotonic	0.970529	0.172450	896

The numerical Brier improvement from sigmoid or beta calibration is only approximately $2 - 3 \times 10^{-6}$, which is operationally negligible. Isotonic regression, meanwhile, reduces the number of unique deployment probabilities to only 141 and creates a boundary tie involving 896 observations. Under pessimistic tie resolution, its NormalizedRecall@5% falls to 0.968282. Raw CatBoost probabilities are therefore retained because they provide essentially equivalent calibration while preserving a stable and finely resolved ranking.

C. Audit Capacity Allocation

Rankings become operational only as constrained portfolios (Table 4.3: 3%, 5%, and 7% capacity).

Table 4.3. Audit allocation performance at 3%, 5%, and 7% capacity

Audit Capacity	Audit Slots	Fraud Captured	Legitimate Audits	Normalized Recall	Lift vs. Random
3%	4,805	4,718	87	0.981894	1.961×
5%	8,008	7,772	236	0.970529	1.938×
7%	11,212	10,761	451	0.959775	1.917×

At the official 5% capacity, the system assigns 8,008 audit slots, of which 7,772 contain fraud-labelled claims and only 236 contain legitimate claims. The resulting portfolio precision is therefore 97.05%.

Random 5% auditing would catch ~4,010 fraud while reviewing ~3,998 legitimate claims; the burden baseline catches 4,844 at 3,164 legitimate audits; the corrected anchor reaches 7,623 at 385. The selected system therefore improves both capture and capacity efficiency.

Expanding 5% to 7% adds 2,989 fraud captures but nearly doubles legitimate audits (236 to 451); shrinking to 3% yields 98.19% precision but leaves more fraud unreviewed - the ranking quantifies this trade-off explicitly.

D. Policyholder Protection

Because the operational cost of false positives is borne by legitimate policyholders, fairness analysis is performed specifically on legitimate claims rather than on the full population. Figure 4.4 presents group-specific audit false-positive rates at the official 5% capacity with 95% Wilson intervals.

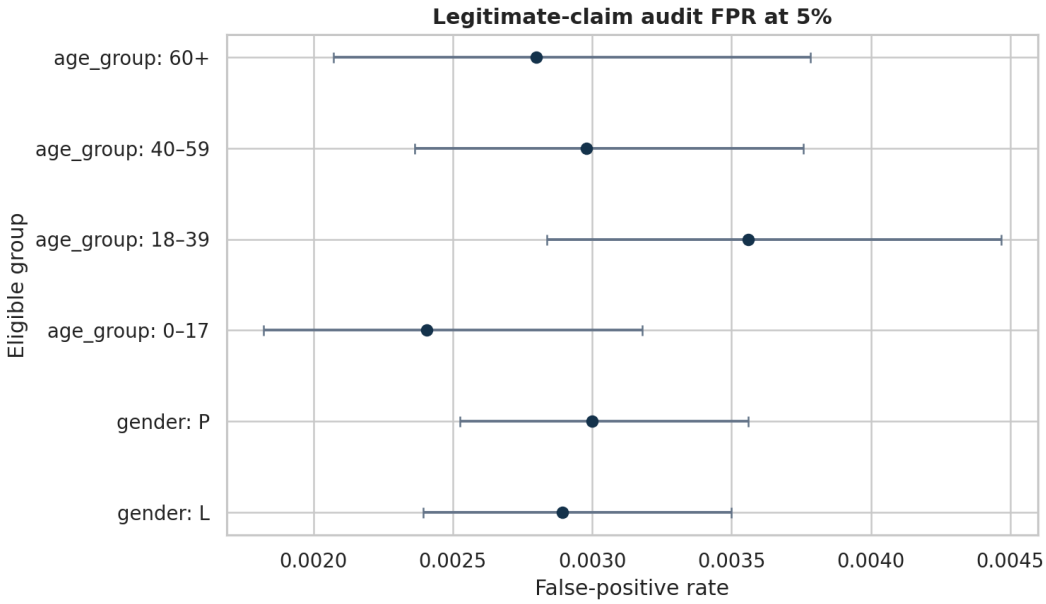


Figure 4.4. Legitimate-claim audit rates at 5% capacity across the required gender and age groups.

At 5% capacity, legitimate-claim audits number 106 of 36,634 legitimate claims for gender L (0.289%) and 130 of 43,337 for gender P (0.300%). By age, 49 of 20,360 (0.241%) for 0–17, 74 of 20,786 (0.356%) for 18–39, 71 of 23,823 (0.298%) for 40–59, and 42 of 15,002 (0.280%) for 60+, confirming the highest exposure in the 18–39 group and the lowest in the 0–17 group.

Gender gap is ~0.0106pp and age gap ~0.1153pp with Protection Score 0.998847; cluster bootstrap gives 0.0009-0.0927pp (gender) and 0.0432-0.2434pp (age).

Highest age rate: 18-39; lowest: 0-17. Differences warrant monitoring, not discrimination conclusions - fraud patterns, utilization, case mix, geography, and data quality all contribute.

A sensitivity screen removing the protected attributes does not automatically improve the protection metric and slightly reduces the overall model evidence. This suggests that simply deleting gender or age is not sufficient to guarantee equal audit exposure, particularly because other clinical or geographic variables may contain correlated information. Direct outcome monitoring is therefore more informative than relying solely on feature exclusion.

E. Model Interpretation and Error Patterns

Strong ranking performance alone is insufficient for an audit-support system. Investigators also need to understand which observed characteristics influence the model and whether these relationships are stable enough to support review.

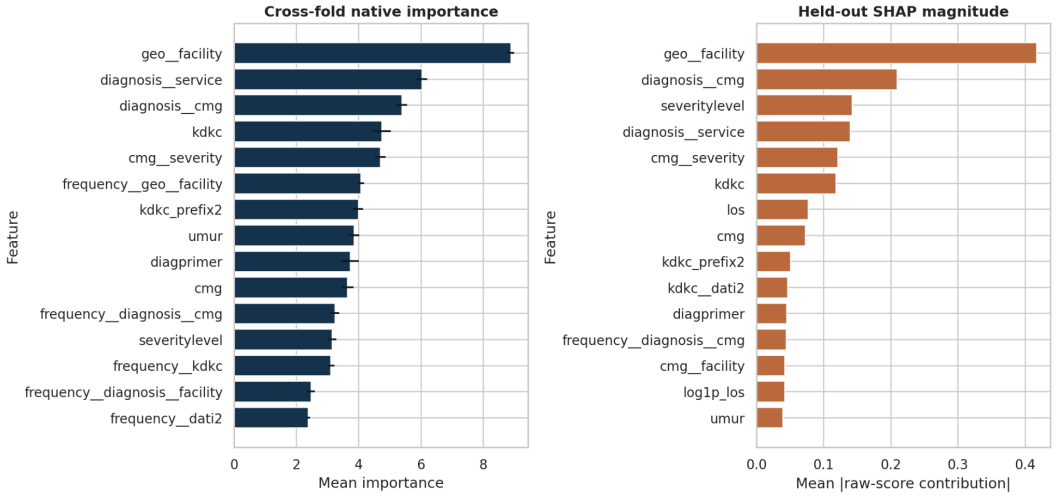


Figure 4.5. Global CatBoost importance and held-out SHAP evidence for the selected model

Across CatBoost native importance and held-out SHAP analysis, the strongest global contribution comes from the interaction between administrative location and healthcare-facility type (geo__facility). Other prominent drivers include diagnosis-service relationships, diagnosis-CMG combinations, service-facility interactions, severity-related structures, geographic hierarchy, and age.

This finding provides two complementary insights. First, fraud risk in the supplied data is not represented by one isolated variable; performance arises from interactions among clinical utilization and administrative context. Second, the prominence of geographic and facility variables requires caution. These variables can represent legitimate differences in provider behavior and healthcare delivery, but may also function as proxies for structural differences between populations. Their importance is therefore interpreted as predictive association rather than evidence that a location, provider category, diagnosis, or demographic characteristic is inherently fraudulent.

Ablating structured interactions, concentration, or rarity features reduces screening scores (interactions most), supporting the compact hypothesis-driven set over indiscriminate expansion.

Per-claim SHAP shows what pushed a score up or down - investigator context, never standalone justification for denial.

F. Robustness and Operational Limitations

The final analysis examines whether strong OOF performance remains stable under alternative conditions.

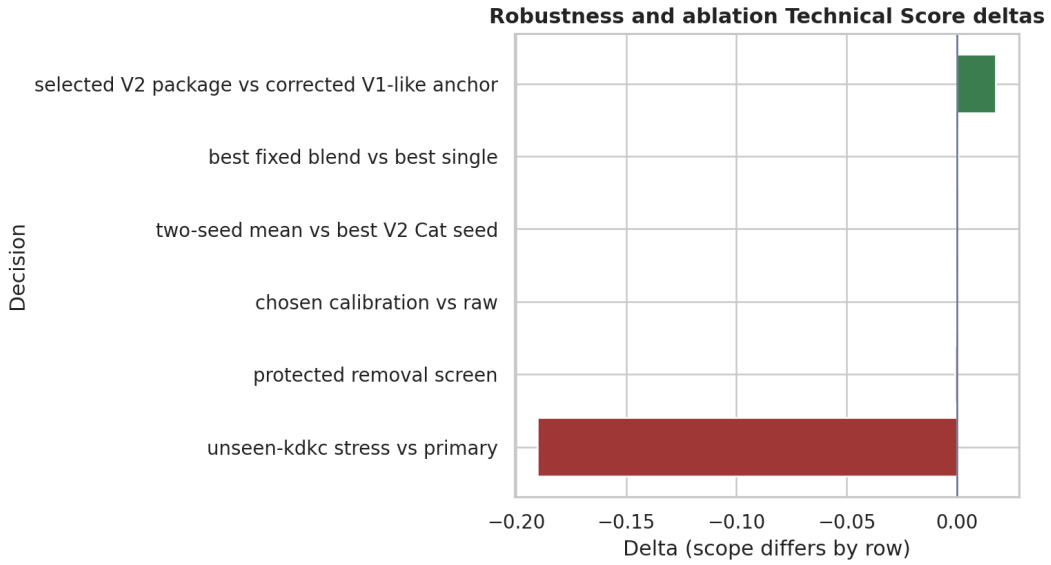


Figure 4.6. Robustness, ablation, and geographic-transport stress results.

Within the official train-test setting, distributional evidence is relatively stable: adversarial validation AUC is approximately 0.5068, and the largest single-feature shift is only 0.0133. However, a substantially different result emerges when entire kdkc geographic codes are withheld from training. Under the standard grouped OOF perspective, CatBoost seed 2026 achieves NormalizedRecall@5% of 0.968032, whereas the unseen-kdkc stress evaluation falls to 0.727937. Average Precision similarly decreases from 0.823088 to 0.609742.

This is the most important robustness limitation identified in the analysis. Geography contains legitimate predictive information within the competition distribution, but the model is less reliable when transported to locations absent from training. The result supports retaining geographic features for the current competition cohort while rejecting the assumption that the same model can be deployed unchanged in a previously unseen region.

Portfolio membership needs monitoring: five fold-models overlap 84.7% pairwise at top-5%, and mean-vs-median probabilities overlap 96.0% - small score gaps still move the boundary. Versioned inference and boundary monitoring are operational requirements.

The selected system is highly effective in-distribution but stays inside a human-centered process monitoring explanations, location context, calibration, subgroup burden, overrides, appeals, and drift (Chapter V).

CHAPTER V: Conclusion and Recommendation

A. Conclusion

This study develops an Integrated Claim Risk and Audit Allocation System for a healthcare-insurance setting where only 5% of incoming claims can be manually reviewed. The analysis shows that reliable fraud prioritization requires more than

model accuracy. Data inspection identified categorical-code inconsistencies, repeated claim signatures, conflicting labels, unusual observations, and a source-order artifact that could create leakage if improperly used. These findings motivate semantic-preserving preprocessing and exact-signature grouped validation.

The selected CatBoost model achieves an out-of-fold NormalizedRecall@5% of 0.97053 and Average Precision of 0.82318. At the official 5% audit capacity, 7,772 of 8,008 audited claims are fraud-labelled, while only 236 legitimate claims are unnecessarily reviewed. Performance remains strong at 3% and 7% capacity, showing that the ranking can support different operational workloads. Calibration testing provides negligible improvement over the raw probabilities, so the original CatBoost scores are retained.

Policyholder-protection analysis shows small observed differences in legitimate-claim audit exposure across gender and age groups, with gaps of approximately 0.011 and 0.115 percentage points, respectively. These disparities should be monitored but should not be interpreted as evidence of discrimination. The main robustness limitation is geographic transportability: performance declines considerably when the model is evaluated on administrative locations absent from training. Therefore, the system is most appropriate as decision support for investigators, not as an autonomous fraud-adjudication mechanism.

Together, these findings answer the research questions: data-quality and leakage risks must first be controlled; an interpretable grouped CatBoost model can provide strong fraud ranking; limited audit capacity can be allocated substantially more efficiently than simple or random strategies; demographic audit exposure can be quantified and monitored; and safe operational use requires human review, explanation, monitoring, and adaptation when data conditions change.

B. Recommendation

Based on these findings, the following operational process is recommended for NHPA.

a. Risk-Based Audit Prioritization
   - Incoming claims should be ranked by fraud risk, with the highest-ranked 5% forming the primary audit queue. Claim verifiers should receive the risk score together with relevant administrative and clinical information and a concise explanation of the main model drivers. The score should indicate investigation priority, not proof that a claim is fraudulent.

b. Human Review and Appeal
   - Every flagged claim should remain subject to human verification. Investigators may override model prioritization when clinical, administrative, or data-quality context provides a documented legitimate explanation. The model must not automatically reject,

suspend, or penalize claims. Disputed decisions should enter an escalation or appeal process reviewed by human decision-makers, with the model output used only as supporting evidence.

c. Performance and Policyholder Monitoring
   - NHPA should routinely monitor fraud yield, legitimate-audit rates, calibration, feature and score distributions, and audit exposure across gender, age, geography, and facility groups. Investigator overrides and appeal outcomes should also be reviewed to identify recurring failure patterns. Where feasible, a small randomized audit sample can help detect emerging fraud patterns that may be systematically ranked too low by the model.

d. Drift, Recalibration, and Retraining
   - Recalibration should be considered when ranking performance remains stable but probability reliability deteriorates. Retraining should be triggered by sustained declines in fraud capture, substantial distribution shift, new geographic or provider populations, recurring missed-fraud patterns, or increasing subgroup disparities. The strong performance decline on unseen locations makes geographic expansion a particularly important trigger for additional validation before deployment.

In practice, the model should prioritize attention, while accountable human reviewers retain authority over investigation and claim decisions.

REFERENCES

[1] A. du Preez, S. Bhattacharya, P. Beling, dan E. Bowen, "Fraud detection in healthcare claims using machine learning: A systematic review," Artif. Intell. Med., 2025, doi: 10.1016/j.artmed.2024.103061.

[2] World Health Organization, Use of Machine Learning for Fraud Detection within the Claims Management Process in the Philippines: Insights from the National Health Insurance Scheme PhilHealth. Geneva, Switzerland: WHO, 2024.

[3] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased boosting with categorical features,” in Advances in Neural Information Processing Systems 31, 2018.

[4] A. Rajkomar, M. Hardt, M. D. Howell, G. Corrado, and M. H. Chin, “Ensuring fairness in machine learning to advance health equity,” Annals of Internal Medicine, vol. 169, no. 12, pp. 866–872, 2018.

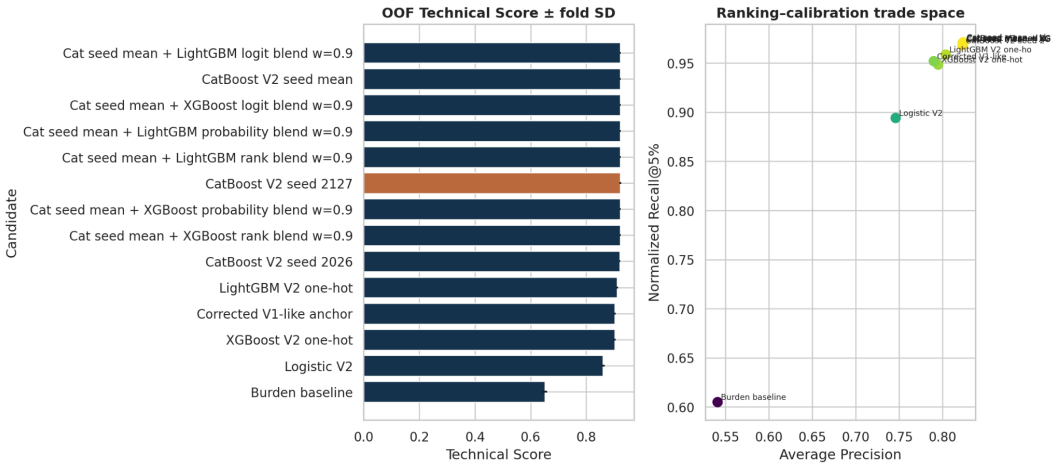
[5] World Health Organization, Ethics and Governance of Artificial Intelligence for Health. Geneva, Switzerland: WHO, 2021.

[6] J. Davis and M. Goadrich, “The relationship between Precision-Recall and ROC curves,” in Proc. 23rd International Conference on Machine Learning, 2006, pp. 233–240, doi: 10.1145/1143844.1143874.

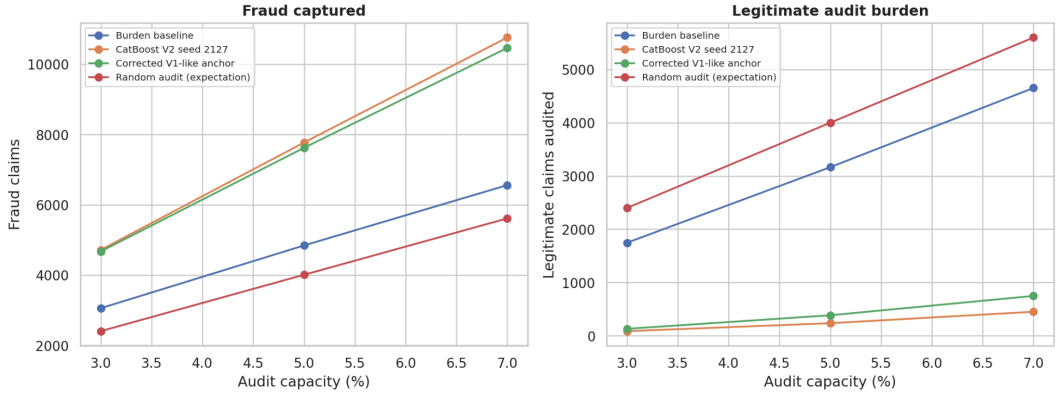
[7] T. Saito and M. Rehmsmeier, "The Precision-Recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLOS ONE, vol. 10, no. 3, Art. no. e0118432, 2015, doi: 10.1371/journal.pone.0118432.
[8] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," ACM Transactions on Knowledge Discovery from Data, vol. 6, no. 4, Art. no. 15, 2012, doi: 10.1145/2382577.2382579.
[9] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," Ecography, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.
[10] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," Journal of Machine Learning Research, vol. 11, pp. 2079–2107, 2010.
[11] G. W. Brier, "Verification of forecasts expressed in terms of probability," Monthly Weather Review, vol. 78, no. 1, pp. 1–3, 1950.
[12] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: The Achilles heel of predictive analytics," BMC Medicine, vol. 17, Art. no. 230, 2019, doi: 10.1186/s12916-019-1466-7.
[13] M. Kull, T. M. Silva Filho, and P. Flach, "Beta calibration: A well-founded and easily implemented improvement on logistic calibration for binary classifiers," in Proc. 20th International Conference on Artificial Intelligence and Statistics, vol. 54, pp. 623–631, 2017.
[14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems 30, 2017.
[15] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," ACM Computing Surveys, vol. 46, no. 4, Art. no. 44, 2014, doi: 10.1145/2523813.

ATTACHMENT

Attachment 1: Common-framework comparison of model ranking performance, stability, and probability quality.



Attachment 2: Fraud captured and legitimate audit burden under 3%, 5%, and 7% audit capacities.



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